Complete Project Documentation: OmniSense-AI

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GitHub Repository: https://github.com/Mehak-Maan/OmniSense-AI
Technology: Multimodal Video RAG, FastAPI, React 19, TypeScript, Groq LLM & Whisper Turbo

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Welcome to the official documentation for OmniSense-AI!
Yeh document aasan aur seedhe alfaaz (easy words) mein likha gaya hai taake koi bhi reader samajh sake ke yeh system kya hai, kaise kaam karta hai, iska pipeline kaisa hai, aur ise apne computer par kaise chalana hai.

1. What is this Project? (Yeh Project Kya Hai?)

OmniSense-AI ek state-of-the-art **Multimodal Video RAG** platform hai. Aam AI chatbots sirf plain text samajhte hain, magar OmniSense-AI ek sath **4 mukhtalif media formats** (Video, PDF Documents, Audio Voice, aur Images) ko analyze karta hai.

RAG ka matlab kya hai? (Retrieval-Augmented Generation):
RAG ka matlab hai: jab aap AI ko koi file (video, PDF ya audio) dete hain, to AI pehle us file mein se sahi information dhoondhta hai (**Retrieval**), aur phir us information ki bunyad par bilkul sach aur accurate jawab likhta hai (**Generation**). Is se AI jhoot ya mann-gharant batein (Hallucination) nahi karta.

Is project se aap kya kar sakte hain?

- **Video Synchronization:** MP4 video upload karein, AI video ke important scenes ko thumbnails ke sath timeline par show karega.
- **Click-to-Jump Video Playback:** Chat mein agar AI kisi second (jaise 00:35) ka zikar kare, to us chip par click karne se video player foran usi second par jump kar jata hai.
- **PDF Document Q&A;:** Financial reports ya technical books upload karein, AI exact page numbers (Page 1, Page 2) cite kar ke jawab dega.
- **Voice & Audio Transcription:** Audio podcast ya meeting recording upload karein, Groq Whisper use seconds mein text mein tabdeel kar dega.
- **Image & Chart Reasoning:** Architecture diagrams, charts, ya dashboards upload karein, Groq Vision unke tamam numbers aur stages ko explain karega.
- **Cross-Modal Attention:** Ek hi waqt mein video, PDF, aur image se mila kar sawal puchen — system screen par show karega ke jawab mein kitna percent hissa video ka tha aur kitna PDF ka.

2. What Technologies & Tools Were Used? (Kaunsi Tech Use Hui?)

Is platform ko industry-standard modern open-source aur cloud AI tools ke sath build kiya gaya hai:

Component	Tool / Library	Role in Project (Aasan Lafzon Mein)
Backend Server	`FastAPI (Python 3.12)`	Super-fast asynchronous REST API jo file upload aur chat requests handle karta hai.

Frontend Web UI	`React 19 + TypeScript + Vite`	Modern, responsive web app jisme video player, timeline cards, aur chat interface hai.
Styling	`Tailwind CSS v4 + Lucide`	Clean dark/light theme, modern cards, aur responsive layouts.
Computer Vision (Video)	`OpenCV (headless)`	Video ko khol kar FPS, duration, aur strategic moments par scene thumbnails banata hai.
Image AI Vision	`Groq Vision (qwen3.8-27b)`	Architecture diagrams, metrics, aur visual keyframes ko decode kar ke markdown banata hai.
Audio Speech-to-Text	`Groq Whisper Turbo`	Audio ya video track ki speech ko word-by-word exact timestamps ke sath transcribe karta hai.
PDF Processing	`PyMuPDF (fitz)`	PDF documents ko page-by-page parhta hai aur ~300 words ke readable chunks banata hai.
Vector Retrieval	`Scikit-Learn (TF-IDF)`	Charon modalities ke chunks ko ek shared vector index me dhal kar <5ms mein dhoondhta hai.
LLM Brain	`Groq LLM (gpt-oss-20b/120b)`	Retrieved context ko parh kar sub-second speed se perfect jawab synthesize karta hai.

3. How Does the Pipeline Work? (Step-by-Step Working)

OmniSense-AI ka data flow 4 scenarios aur 1 core retrieval engine par mushtamil hai:

Scenario A: When You Upload a Video (Video Ingestion Pipeline)

1. **Video Inspection:** OpenCV video file open karta hai aur duration, resolution (e.g. 1920x1080), aur FPS calculate karta hai.
2. **Scene Keyframing:** Video ke dynamic intervals (10%, 35%, 65%, 90%) se high-quality frames capture karta hai aur 320x180 thumbnails save karta hai.
3. **Vision Scene Captioning:** Groq Vision har keyframe ko inspect kar ke 2–4 words ka title aur scene description generate karta hai.
4. **Audio Extraction:** FFmpeg video se audio track (16kHz PCM WAV) alag karta hai aur Groq Whisper ko bhej kar poori speech transcript hasil karta hai.
5. **Indexing:** Har video scene ko uske exact timestamp (e.g. 00:35) ke sath RAG Vector Index mein store kar diya jata hai.

Scenario B: When You Upload a PDF Document

1. **Parsing:** PyMuPDF PDF ko page-by-page parhta hai.
2. **Chunking:** Har page ke text ko 300-word ke logical paragraphs mein break karta hai.
3. **Page Tagging:** Har chunk ke sath file ka naam aur Page Number (Page 1, Page 2) tag hota hai.
4. **Vector Storing:** Paragraphs RAG vector store mein index ho jate hain.

Scenario C: When You Upload an Image (Diagrams / Charts / Screenshots)

1. **Base64 Encoding:** Image file ko base64 format mein convert kiya jata hai.
2. **Vision OCR & Layout Analysis:** Groq Vision image ke headers, 4 KPI cards, flowcharts, tables, aur security boxes ko markdown text mein extract karta hai.
3. **Section Breakdown:** Markdown headers (##) ke zariye chote chunks bana kar index karta hai taake diagram ka koi bhi number foran search ho sake.

Scenario D: When You Ask a Question (The RAG Retrieval & Synthesis Pipeline)

1. **Query Vectorization:** User ka sawal TF-IDF model ke zariye feature vector mein tabdeel hota hai.
2. **Cosine Similarity Matching:** Vector index charon modalities (Video + PDF + Audio + Image) ke saare chunks ko compare kar ke **Top-6 sab se relevant** chunks select karta hai.
3. **Dynamic Attention Weights:** Har modality ke similarity scores ko jod kar percentage calculate ki jati hai (e.g., Video: 56%, PDF: 31%, Image: 13%, Audio: 0%).
4. **Timestamp Extraction:** Agar chunk video ya audio ka ho, to exact second offsets (00:35, 01:30) extract ho kar clickable chips ban jate hain.
5. **Groq LLM Synthesis:** Grounded context Groq LLM (gpt-oss-20b/120b) ko diya jata hai, jo sub-second speed mein accurate, structured aur cited answer generate karta hai.

4. Project Folder Structure Explained (Files Ki Tafseel)

Repository ke andar har file aur folder ka ek wazeh maqsad hai:

File / Folder	Purpose & Functionality (Kyun Aur Kya Hai)
`backend/main.py`	Main FastAPI application. Isme OpenCV video processing, PyMuPDF parsing, Whisper transcription, TF-IDF vector index, aur Groq LLM chat engine implement hain.
`backend/requirements.txt`	Backend dependencies list: fastapi, uvicorn, opencv, pymupdf, groq, scikit-learn, pillow, reportlab.
`backend/.env.example`	API keys ki template file. Real keys public nahi hoti, user ise copy kar ke apni keys add karta hai.
`frontend/src/App.tsx`	Main React application. Video player, dynamic timeline scrubbers, attention weight bars, drag-and-drop file uploaders, aur real-time chat interface contain karta hai.
`frontend/package.json`	Frontend React dependencies: Vite, Tailwind CSS v4, Lucide-react icons, React-Markdown.
`sample_videos/`	High quality sample MP4 video testing ke liye (Multimodal AI tutorial).
`sample_pdfs/`	Financial & enterprise report PDF with tables and metrics for document Q&A; testing.
`sample_audios/`	Product strategy meeting recording for Whisper speech recognition testing.

`sample_images/`	System architecture dashboard image for visual OCR & diagram reasoning.
`README.md`	Full GitHub landing documentation with Mermaid system diagrams, API specs, and run guides.

5. Easy Setup Guide (Apne Computer Par Kaise Chalayein)

Is project ko run karna intehai aasan hai. Sirf yeh simple steps follow karein:

```
# Step 1: Clone the GitHub Repository
git clone https://github.com/Mehak-Maan/OmniSense-AI.git
cd OmniSense-AI

# Step 2: Set Up & Run the Backend
cd backend
python -m venv venv
.\venv\Scripts\activate # (Linux/Mac: source venv/bin/activate)
pip install -r requirements.txt
# Create .env and put your GROQ_API_KEY=gsk_...
uvicorn main:app --host 0.0.0.0 --port 8000 --reload
Backend live ho jayega: http://localhost:8000

# Step 3: Set Up & Run the Frontend (New Terminal)
cd frontend
npm install
npm run dev
Frontend live ho jayega: http://localhost:5173

# Step 4: Open Browser & Test
Browser mein http://localhost:5173 open karein aur top bar mein 'Load Demo Video & Data' par click
karein. AI engine foran chal padega!
```

Official GitHub Repository & Project Link:
Aap is project ka mukammal code, frontend, backend, aur sample assets yahan dekh sakte hain:
https://github.com/Mehak-Maan/OmniSense-AI

Main Features Summary:
• 4 Modalities: Video (.mp4), PDF (.pdf), Audio (.wav/.mp3), Image (.png/.jpg)
• Synchronized Video Player with click-to-jump bookmarks (00:35, 01:30)
• Dynamic Modality Attention Distribution (Video %, PDF %, Image %, Audio %)
• Sub-Second Groq LLM Inference with Zero Hallucination Grounding